

H2GUARD · DESIGN DECISIONS REGISTER

H2Guard design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2guard		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 12 (HGD-DDR-003 accepted; head on a ceiling drop rod); moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions gathered from the review note and the decision records; budget treated as a value-engineering target

H2Guard design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The catalytic sensor's response time (t90), power and hydrogen output, from its datasheet	R4 assumes a t90 of 30 s; R1's 1 % LFL resolution needs the output	HGD-CAL-001, F3
2	The sensor can is about 20 mm across and 17 mm tall	The standoff length sets its 2.2 mm gap above the disc	HGD-DDR-003, P4
3	The head box has its lid on a 110 x 90 face	The ports, gland and wall holes assume it	HGD-DDR-003, P3
4	The controller box's boss spacing and corner fixing holes	They set the mounting plate holes and the wall screws	HGD-DDR-003, P6, P8
5	The fan's body fits a 200 mm sleeve (about 180 mm across or less) and has fixing holes in its inlet face	The fan plate holds it by those holes	HGD-DDR-003, P10
6	The fan's curve gives at least 300 m ³ /h against about 60 Pa	R7 rests on an assumed curve	HGD-CAL-001, G3
7	The valve has two tapped holes underneath, 36 mm apart, and is rated for hydrogen with a stated seat leak rate	The bracket holes; the leaking-seat fault in Table 4 of HGD-CAL-001	HGD-DDR-003, P11; HGD-DDR-001, O5

■ Value engineering

Value-engineering target: USD 265 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 289 (USD 24 over the target), from the 16-line bill of materials at indicative prices. Main cost drivers and savings worth trying:

- The largest lines are the exhaust fan with its grille, sleeve, shutter and hood (USD 55), the catalytic sensor (USD 35), the solenoid valve (USD 30) and the controller modules (USD 25). Together they are about half the cost.
- Making the design constructable added USD 25: the made brackets and plates (USD 14), the 200 mm wall sleeve (USD 5), standoffs, screws and epoxy (USD 4) and the cup's push-in fitting and tube clips (USD 2).
- Decided on 2026-10-02 and not yet priced: the ceiling drop rod for the head, the differential pressure switch, the dry-contact relay output, the clear-lid controller box, a hydrogen-rated valve if the brass valve fails its proof, and the oxygen sensor option where inert gas cylinders share the room. They will raise the estimate.

- Savings worth trying: a printed controller mounting plate in place of aluminium sheet (about USD 3); buying the fan as a kit with its own shutter and hood; one combined controller board at TRL 4 in place of separate bought modules; and, where the room already has mechanical extract at high level, using it in place of the kit fan and grille (USD 63), which would bring the kit to about USD 226.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D9: catalytic plus metal oxide sensors (certified sensor as the upgrade); set points 10 % and 25 % LFL; normally closed valve; continuous ventilation with boost; inventory rule (R9); key-switch reset and one head per room; a university teaching lab first; independent hardware trip; 24 V DC with a certified supply	Amish: "i accept all your recommendations, go with them across all repos."	HGD-DDR-001, HGD-DDR-002
2026-09-25	Budget figure USD 265, now treated as the value-engineering target	Amish, same instruction; on 2026-10-01: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	HGD-DDR-002
2026-09-25	Timed escalation: a warning held 5 min closes the valve and latches	Amish, same instruction	HGD-DDR-002
2026-09-25	Head placement rule: a head directly above each likely leak point; offset limit set by test at TRL 4	Amish, same instruction	HGD-DDR-002
2026-09-25	R13 scope: the fan and grille wall openings are builder's work	Amish, same instruction	HGD-DDR-002
2026-09-25	H2Bench to reflect the inventory rule	Amish, same instruction	HGD-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan; keep open decisions out of the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	HGD-DDR-003 (changes accepted on 2026-10-02, below)
2026-10-02	Design for construction accepted: the changes P1 to P11, as made; the head position is settled separately (decision 2)	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-003, Table 1
2026-10-02	The head hangs from the ceiling on a drop rod directly over the apparatus; wall mounting is allowed only if the TRL 4 plume test shows a trip at the wall position (the wall position puts the ports 260 mm from the apparatus centre against a plume radius of about 141 mm)	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-003, A1
2026-10-02	Fan motor in the sleeve accepted for the supervised teaching prototype; any room used without close supervision gets a fan with its motor outside the air stream	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-003, A2
2026-10-02	An oxygen sensor is added, as a standard kit option, whenever inert gas cylinders share the room	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-001, O2
2026-10-02	Set points and the inventory rule come from the stricter of the national code and ISO 26142, and from the host institution's safety office where its	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-001, O3

Date	Decision	Decided by	Record
	rules are stricter; 10 % and 25 % LFL stay the defaults		
2026-10-02	A differential pressure switch at the fan proves airflow	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-001, O4
2026-10-02	The valve is rated for hydrogen with a stated seat leak rate; the low-cost brass valve is accepted only if its maker states hydrogen compatibility and a seat leak rating and it passes a seat leak proof test at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-001, O5
2026-10-02	The 1 % full-span bump test trips the system, as planned; the pass level is set once the cup delivery factor is measured	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-25
2026-10-02	A dry-contact relay output that opens on a trip or loss of power is added, for H2Bench to cut its own supply	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-25
2026-10-02	The render layout is kept, with a caption saying the head and valve are drawn beside the controller	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 1
2026-10-02	A clear-lid IP65 controller box, added to the BOM at its next revision	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 2
2026-10-02	Leaving the fan, grille, sounder, beacon and supply out of the product renders accepted	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 3